Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/29/2006
PI Name:	Love, Nancy Ph.D.		
Project Title:	Integrated Biotreatment Technology for Nitrogen-Rich Wastewaters in Advanced Life Support Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Virginia Tech		
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Zip Code:	24061	**Congressional District:**	9
Comments:	The address above is the address I had when I completed the work. Virginia Tech remains the responsible institution. My current address is: Dept of Civil and Environmental Engineering, University of Michigan, 2340 GG Brown, 2350 Hayward Street, Ann Arbor, MI 48109		
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	09/23/2004	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ04HD78G		
Performance Goal No.:			
Performance Goal Text:			
	To develop a functional, closed-loop water recovery system for Space Shuttle and International Space Station operations, technologies are required to treat blended urine and humidity condensate wastewaters, which contain high concentrations of reduced nitrogen (~ 1.6 g/L as N). Removing this N via conventional treatment means is both resource and cost intensive. This project proposes to develop, test, and optimize a novel, less resource-intensive, microgravity-compatible, high-capacity nitrogen removal technology termed the oxygen-limited autotrophic nitrification-denitrification (OLAND) process. This process, successfully demonstrated by this team, offers the advantages of effective nitrogen removal to dinitrogen gas (N2), no reliance on external electron donors, and low biomass generation. OLAND will be performed within a hollow fiber membrane bioreactor (HFMBR) which provides ample support for biofilm growth, permits bubbleless gas addition, and enables very high volumetric removal rates. It is		

Task Description:	anticipated that at least 90% of the total Kjeldahl nitrogen in the wastewater will be converted to N2 with the remaining exiting as nitrate-N; this can be removed downstream via biological or physico-chemical polishing. An OLAND-HFMBR will be constructed and assessed using organic carbon-free synthetic wastewater during the first phase of the project. Experimentally-determined kinetic and stoichiometric parameters and biofilm community structure information will be used to develop computational models for system performance. These models will provide a basis for subsequent system optimization and scale-up. Also, an anaerobic (organic carbon) stabilization reactor will be installed upstream of the OLAND-HFMBR to determine the impact of the residual organic carbon on the performance of the OLAND-HFMBR. The data gathered during this ground-based study will lay the framework for advancing this process to higher Technology Readiness Levels. The long-term goal of this project is to provide a science-based, robust, space- and microgravity-compatible, biotechnological approach to treat wastewaters generated in Shuttle and ISS operations. Removing nitrogen from these highly nitrogenous wastewaters through the traditional biological paradigms would be prohibitively resource-intensive. The immediate objective of our project is to develop, optimize, and demonstrate a novel approach – based on HFMBR characteristics and the ability for simultaneous aerobic and anaerobic autotrophic ammonium oxidation metabolism in oxygen limited biofilms (OLAND) – to remove nitrogenous substances from this waste stream. This combination of the successfully demonstrated OLAND technology within a HFMBR configuration permits increased process control. This technology represents one essential component of an integrated waste treatment operation for ALS. The specific objectives of this work are to: 1. Design, construct and evaluate the autotrophic nitrogen removal system using synthetic urine/humidity condensate devoid of carbonaceous matter. 2. Develop and assess quantitative process/design models for the autotrophic ammonium removal reactor (OLAND-HFMBR) using a two-step process. 3. Introduce anaerobic organic matter stabilization reactor upstream of autotrophic nitrogen removal process. This project will yield an inexpensive and effective method for nitrogen removal from blended urine and humidity condensate wastewaters generated in Space Shuttle transit and International Space Station applications. The use of the proposed staged biological processing prior to physical-chemical polishing will dramatically reduce the costs associated with regeneration or disposal of physical components (reverse osmosis membranes) or chemical treatment. The use of HFMBRs for treatment of this wastewater offers the advantages of bubbleless gas transfer, development of a redox-stratified biofilm for completely autotrophic nitrogen removal, and decreased operational costs that might otherwise be associated with addition of an external carbon source or electron donor. The system will be built to be compatible with constraints imposed by a microgravity environment. The successful demonstration of this technology in a ground-based research project paves the way for further testing and validation in simulated microgravity environments on the ground as well as testing and demonstration of a prototype unit in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The technology has application to similar high ammonia waste systems on Earth, such as agricultural wastes, domestic sludge processing waters, etc.
Task Progress:	Funding for the project arrived September 23, 2004. Because of the lateness of the funding, the project was not started until January 2005 when the first Ph.D. student could be recruited. Since that time, the following tasks have been performed: 1. An anammox culture was obtained from Dr. Willy Verstraete in Belgium and used to inoculate a preliminary biofilm reactor. The reactor is still running and is being monitored for anammox stoichiometry. 2. Communcations were initiated with Jay Garland at Kennedy Space Center, who showed us how to design a hollow fiber membrane reactor. The reactor has been constructed and it now running. 3. Ersatz wastewater has been synthesized and its treatment by anaerobic digestion was determined. A paper summarizing the outcomes from that work is being presented July 2006 at the SAE conference. 4. The modeling effort is well underway and an initial model has been developed. Initial sensitivity analysis work is ongoing. 5. An anammox culture is being maintained and will be used to inoculate a second reactor system as a backup to the hollow fiber membrane system.
Bibliography Type:	Description: (Last Updated: 03/23/2017)
Papers from Meeting Proceedings	Gilmore, K. R.; Capuno, R. E.; Love, N. G.; and Smets, B. F. "Anaerobic Stabilization of Early Planetary Base Ersatz Wastewater Formulation." SAE - International Conference on Environmental Systems, July 2006. Paper number 06ICES-177, July 2006. , Jul-2006
Presentation	Capuno, R. E.; Love, N. G.; and Smets, B. F. "Mathematical Modeling of Nitrogen Removal via Partial Nitrification and Anaerobic Ammonia Oxidation Biofilm in a Hollow Fiber Membrane Bioreactor" Virginia Water Environment Association Annual Meeting May-2006